# LESSON PLAN

## ROCKET ENGINE (17)

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| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 09 |
| **Total No. of Lesson Plans** | 9 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

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|---|---|
| **1. Topic** | Advanced Composite i.e. carbon-carbon composite, metal-matrix composite and their utilization |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Advanced composite i.e. carbon-carbon composite, metal-matrix composite and their utilization. |
| **5. Specific Objectives** | 17.1.1.1 Trainee will be able to understand the development of composite materials, types and functions of fibers and resin, Defects in composite, selection criteria for Matrix, fabric weaves, prepress, strength flowcharts |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

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| **7. Practical in Lab/Equipment Rooms/Simulators** | Examination of Carbon-Carbon composite samples for defects; Instructor supervision required for safety. |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

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| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What are the primary components of a composite material?
(b) Name one application of carbon-carbon composites in aerospace.
(c) Why is matrix selection critical in composite design?

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Development of composite materials<br>• Types and functions of fibers<br>• Functions of resin matrix<br>• Fabric weaves and prepress | L & Ds | PPT, IB | 15 min |
| (b) | • Defects in composite materials<br>• Selection criteria for Matrix<br>• Strength flowcharts analysis<br>• Utilization of advanced composites | L & Ds | PPT, PDF | 20 min |
| (c) | • Metal-matrix composite properties<br>• Carbon-carbon composite applications<br>• Comparison of matrix types<br>• Final review of selection criteria | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain the role of the matrix in a metal-matrix composite.
(b) Identify two common defects found in carbon-carbon composites.
(c) How does fabric weave affect the strength of a composite?

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| **12. Home Assignment and Follow Up** | Prepare a brief note on the utilization of carbon-carbon composites in hypersonic vehicle applications for the next lesson. |
| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | End of subject | Refer to the master syllabus reference list | - |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
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**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min